Bài 7 – Phân tích và Quản lý đầu tư nâng cao

Markowitz Portfolio Optimization and Capital Asset Pricing Model

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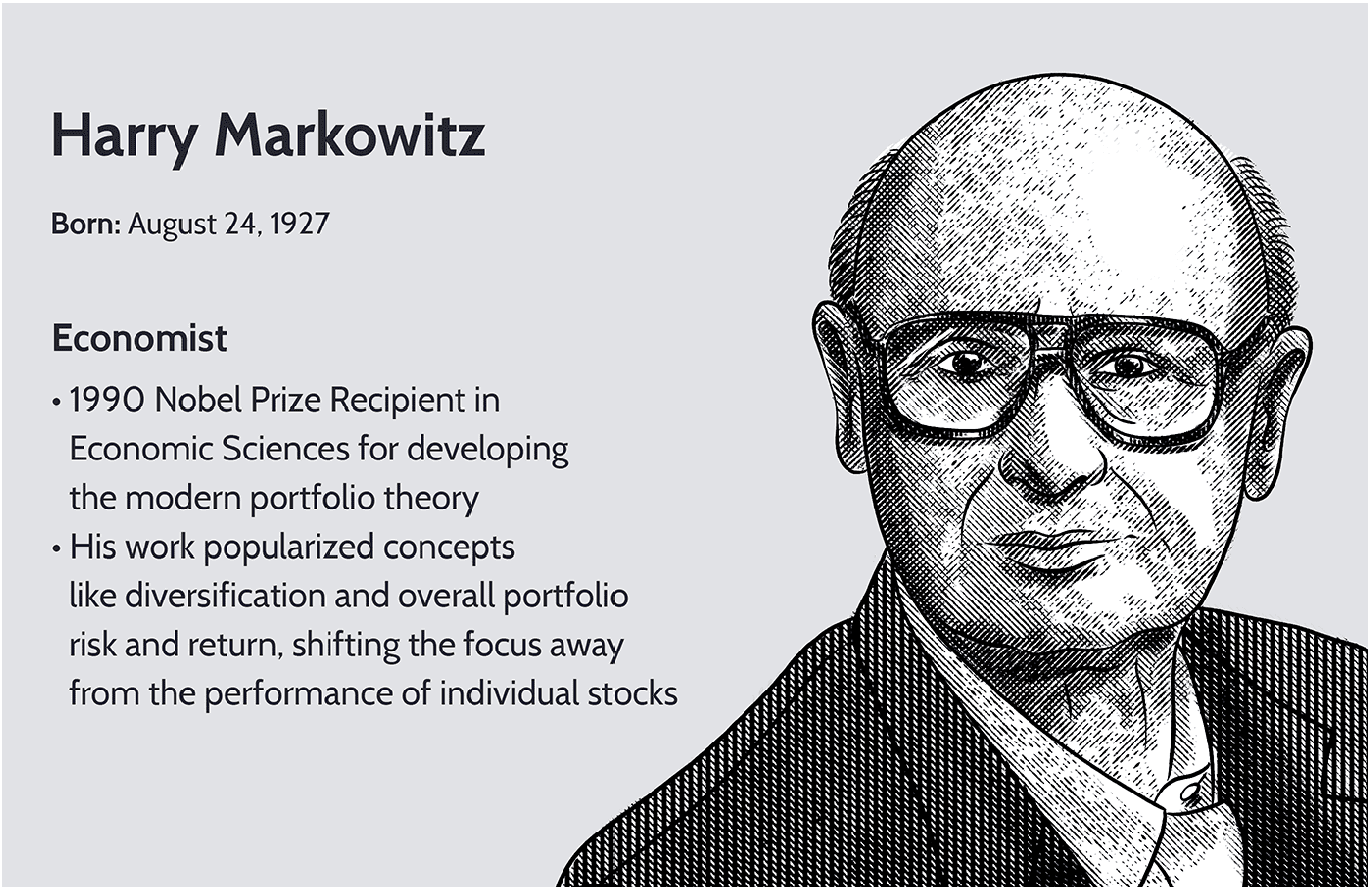
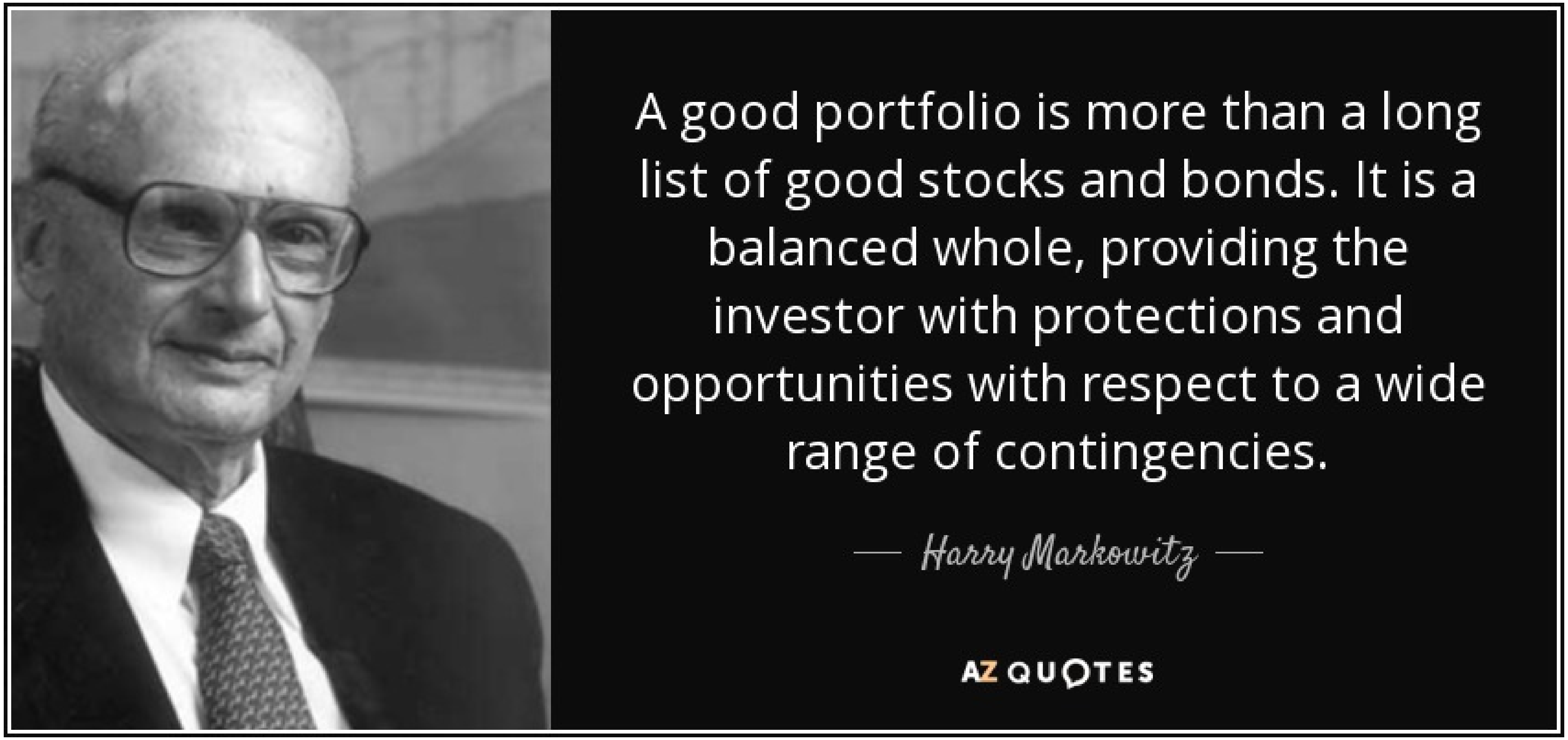
Portfolio Theory

Modern Portfolio Theory

- In 1952, Harry Markowitz published one of the most important papers in the world finance.



Modern Portfolio Theory



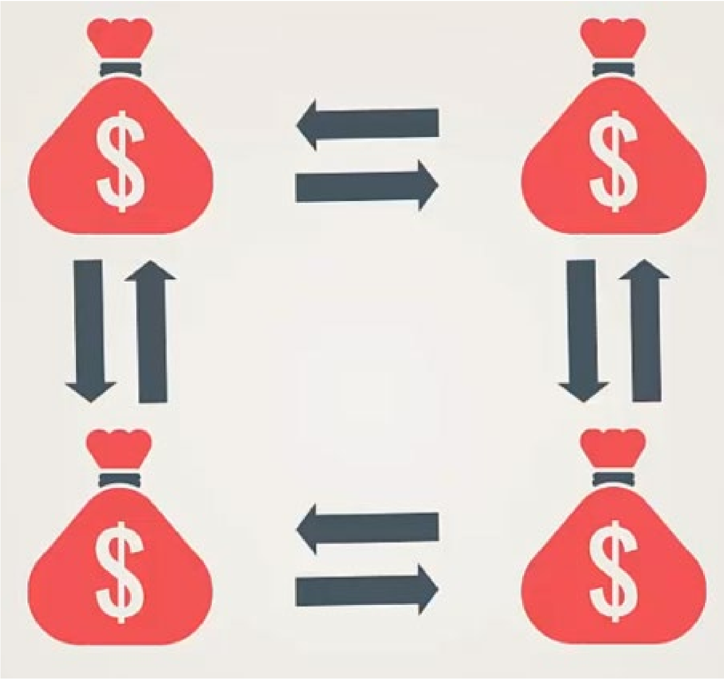
Modern Portfolio Theory

- Markowitz proved the existence of an efficient set of portfolios

- Investors should understand the relationship between securits in their portfolio

Markowitz Portfolio Theory

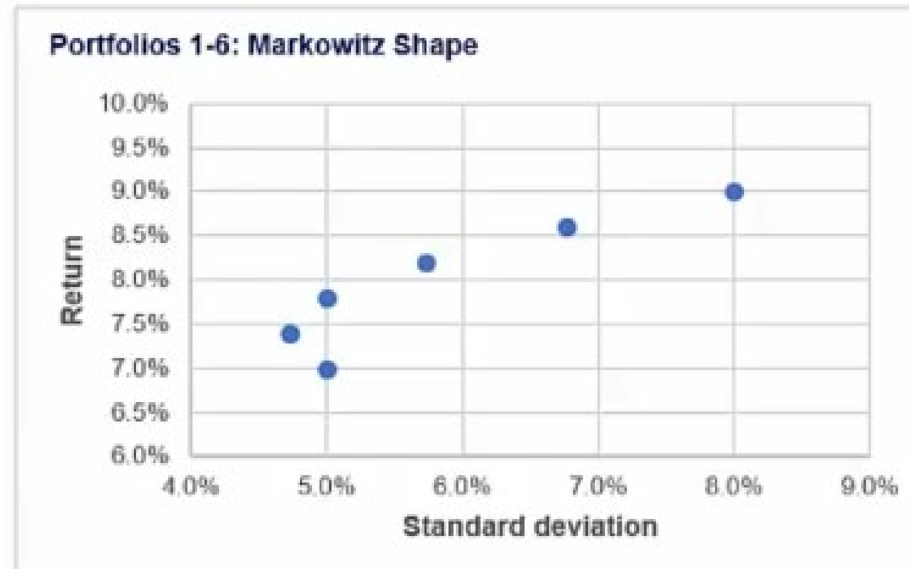
- The combination of securities with little correlation allows investors to optimize their return without assuming additional risk



Calculation

	Share A	Share B
Expected return	7%	9%
Standard deviation	5%	8%
Correlation	30%	

	Weight A	Weight B	Expected Return	Standard Deviation
Portfolio 1	100%	0%	7.00%	5%
Portfolio 2	80%	20%	7.40%	5%
Portfolio 3	60%	40%	7.80%	5%
Portfolio 4	40%	60%	8.20%	6%
Portfolio 5	20%	80%	8.60%	7%
Portfolio 6	0%	100%	9.00%	8%



$$\boxed{w_1 r_1 + w_2 r_2 = r_p}$$

$$\boxed{(w_1\sigma_1 + w_2\sigma_2)^2 = w_1^2\sigma_1^2 + 2w_1\sigma_1 w_2\sigma_2\rho_{12} + w_2^2\sigma_2^2}$$

Quiz

According to Harry Markowitz, investments in multiple securities:

- Should be studied as a portfolio (correct)
- Must be analyzed separately
- Should be done only by mid- and large-sized companies
- Are supposed to increase the expected total return

Pratice

Lab06_01_Obtaining the Efficient Frontier_1

```
import numpy as np
import pandas as pd
from pandas_datareader import data as wb
import matplotlib.pyplot as plt
import yfinance as yf

yf.pdr_override()
%matplotlib inline
```

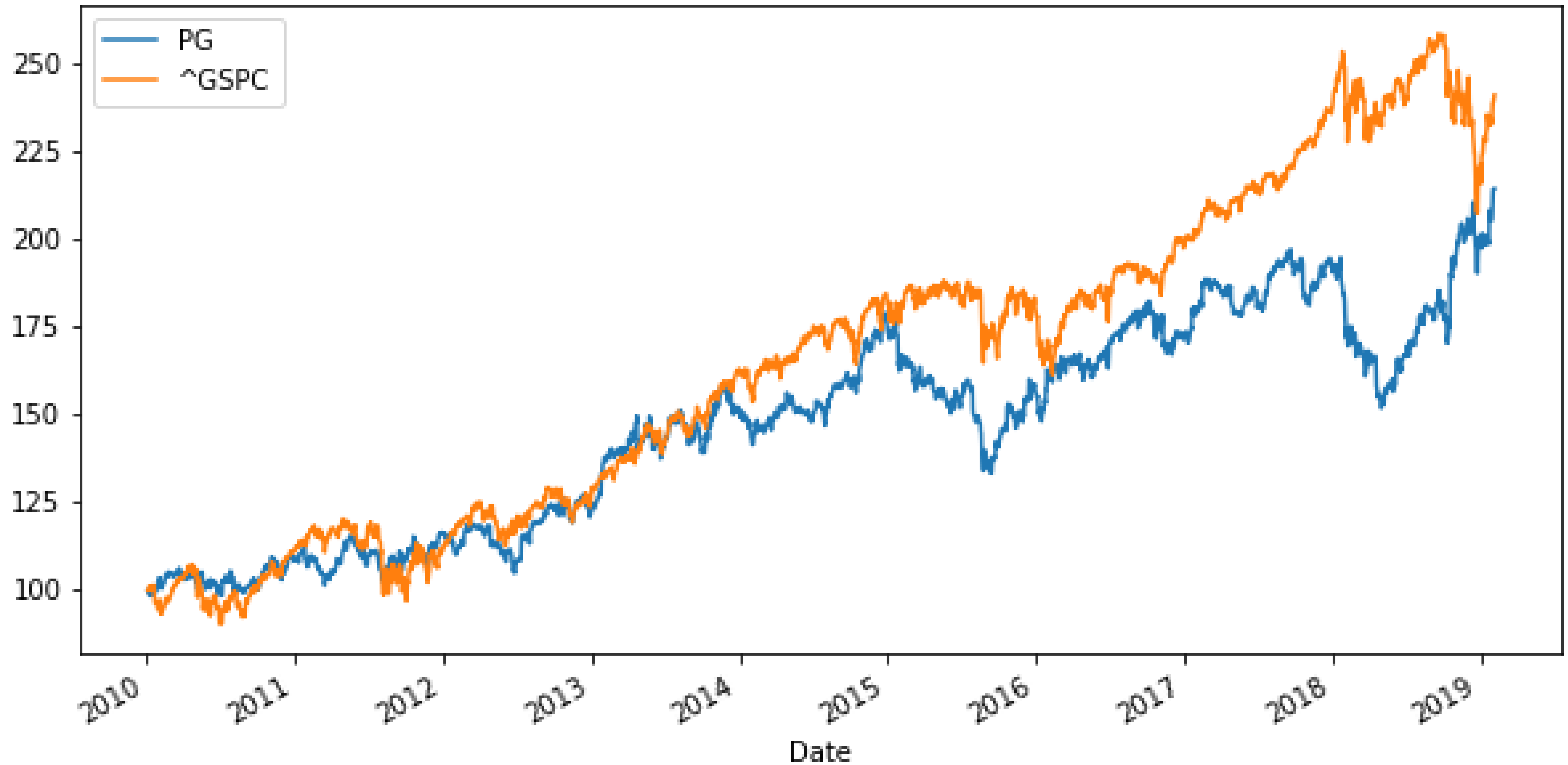
```
[ ] assets = ['PG', '^GSPC']
    pf_data = pd.DataFrame()

    for a in assets:
        pf_data[a] = wb.DataReader(a, start = '2010-1-1')['Adj Close']
```

```
[ ] pf_data.tail()
```

	PG	^GSPC
Date		
2019-01-29	93.540001	2640.000000
2019-01-30	94.519997	2681.050049
2019-01-31	96.470001	2704.100098
2019-02-01	97.470001	2706.530029
2019-02-05	97.639999	2733.030029

```
[ ]  (pf_data / pf_data.iloc[0] * 100).plot(figsize=(10, 5))
```

<matplotlib.axes._subplots.AxesSubplot at 0x16b28ba91d0>



```
[ ] log_returns = np.log(pf_data / pf_data.shift(1))
```

```
[ ] log_returns.mean() * 250
```

```
PG       0.083479
^GSPC    0.096298
dtype: float64
```

```
[ ] log_returns.cov() * 250
```

	PG	^GSPC
PG	0.021374	0.011683
^GSPC	0.011683	0.022498

```
[ ] log_returns.corr()
```

	PG	^GSPC
PG	1.000000	0.532774
^GSPC	0.532774	1.000000

```
[ ] num_assets = len(assets)
```

```
[ ] num_assets
```

```
2
```

```
[ ] arr = np.random.random(2)
    arr
```

```
array([0.73707909, 0.80143814])
```

```
[ ] arr[0] + arr[1]
```

```
1.5385172241835559
```

```
[ ] weights = np.random.random(num_assets)
    weights /= np.sum(weights)
    weights
```

```
array([0.59670321, 0.40329679])
```

```
[ ] weights[0] + weights[1]
```

```
1.0
```

Capital Asset Pricing Model (CAPM)

CAPM Setting

- Investors are:

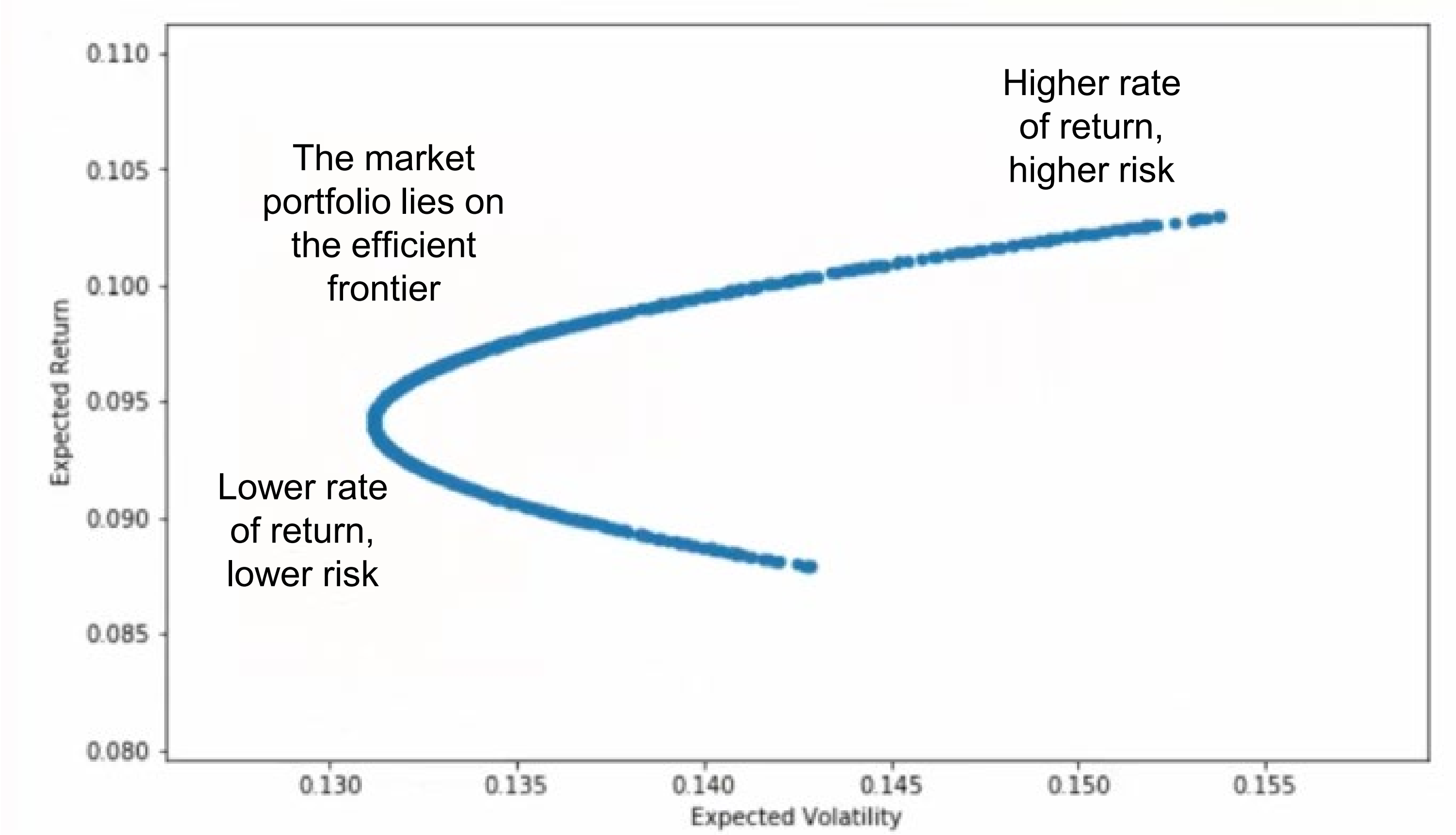
- Using CAPM, investors will invest in:

Depending on their risk preferences, they will choose to buy more of the risk-free asset or more of the market portfolio.

Market Portfolio

- A combination of all possible investments in the world

- Efficient frontier



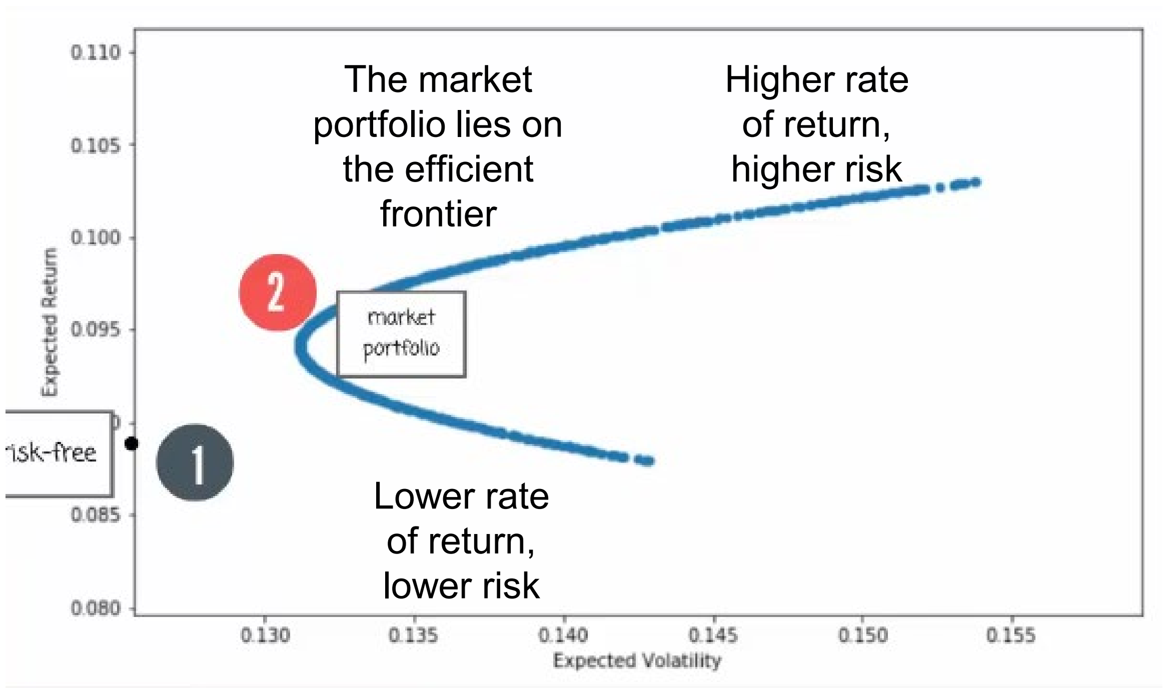
Risk-free Asset

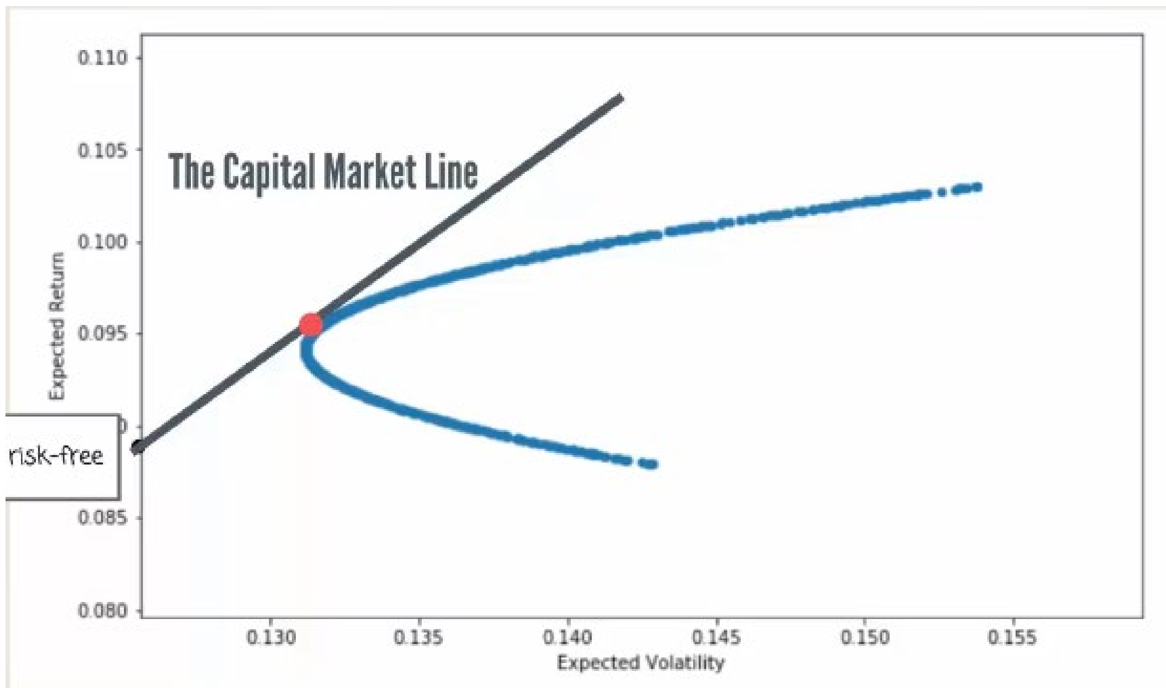
- CAPM assumes the existence of a risk-free asset → An investment with zero risk.
- Why should we assume the risk-fre rate has a lower expetected rate of return?

In efficient markets, investors are only compensated for the added risk they are willing to bear.

Efficient frontier

- Investors will allocate their money between the risk-free and the market portfolio





- The point where the capital market line intersects the efficient frontier is the market portfolio

Depending on their risk preferences, they will choose to buy more of the risk-free asset or more of the market portfolio.

Quiz

In the Capital Asset Pricing Model (CAPM), investors will:

- Always buy the risk-free asset
- Always buy the market portfolio
- Buy more of the risk-free asset or more of the market portfolio, depending on their risk preferences (correct)
- Buy the portfolio with the lowest risk

Pratice

Lab06_02_Obtaining the Efficient Frontier_2

Expected Portfolio Return:

```
[ ]  np.sum(weights * log_returns.mean()) * 250
```

```
0.08644189435560486
```

Expected Portfolio Variance:

```
[ ]  np.dot(weights.T, np.dot(log_returns.cov() * 250, weights))
```

```
0.017977305454256277
```

Expected Portfolio Volatility:

```
[ ]  np.sqrt(np.dot(weights.T,np.dot(log_returns.cov() * 250, weights)))
```

```
0.13407947439580853
```

```
[ ] pfolio_returns = []
    pfolio_volatilities = []

    for x in range (1000):
        weights = np.random.random(num_assets)
        weights /= np.sum(weights)
        pfolio_returns.append(np.sum(weights * log_returns.mean()) * 250)
        pfolio_volatilities.append(np.sqrt(np.dot(weights.T,np.dot(log_returns.cov() * 250, weights))))

    pfolio_returns, pfolio_volatilities
```

```
([0.08445973848769973,
  0.09465311419904361,
  0.08688519664254303,
  0.08492069610542369,
  0.08905148959314078,
  0.09041971777048222,
  0.08624039343860482,
  0.09478544869454558,
```

```
[ ] pfolio_returns = []
    pfolio_volatilities = []

    for x in range (1000):
        weights = np.random.random(num_assets)
        weights /= np.sum(weights)
        pfolio_returns.append(np.sum(weights * log_returns.mean()) * 250)
        pfolio_volatilities.append(np.sqrt(np.dot(weights.T,np.dot(log_returns.cov() * 250, weights))))

    pfolio_returns = np.array(pfolio_returns)
    pfolio_volatilities = np.array(pfolio_volatilities)

    pfolio_returns, pfolio_volatilities
```

```
(array([0.08940066, 0.08938807, 0.09313669, 0.08963229, 0.09261475,
        0.08778874, 0.08928526, 0.0859052 , 0.09035742, 0.08659252,
        0.09293394, 0.09145331, 0.08736617, 0.09014068, 0.08790625,
        0.0898365 , 0.09080135, 0.08902164, 0.09560093, 0.09308266,
        0.08443123, 0.09457634, 0.09473482, 0.09007522, 0.09148244,
        0.08894662, 0.08779015, 0.08679505, 0.09053612, 0.08351064,
        0.09091384, 0.08948242, 0.08834259, 0.09035382, 0.09361975,
        0.08914967, 0.08916253, 0.09008063, 0.08665393, 0.0901195 ,
```

Pratice

Lab06_0303_Obtaining the Efficient Frontier_3

```
portfolios = pd.DataFrame({'Return': pfolio_returns, 'Volatility': pfolio_volatilities})
```

```
[ ] portfolios.head()
```

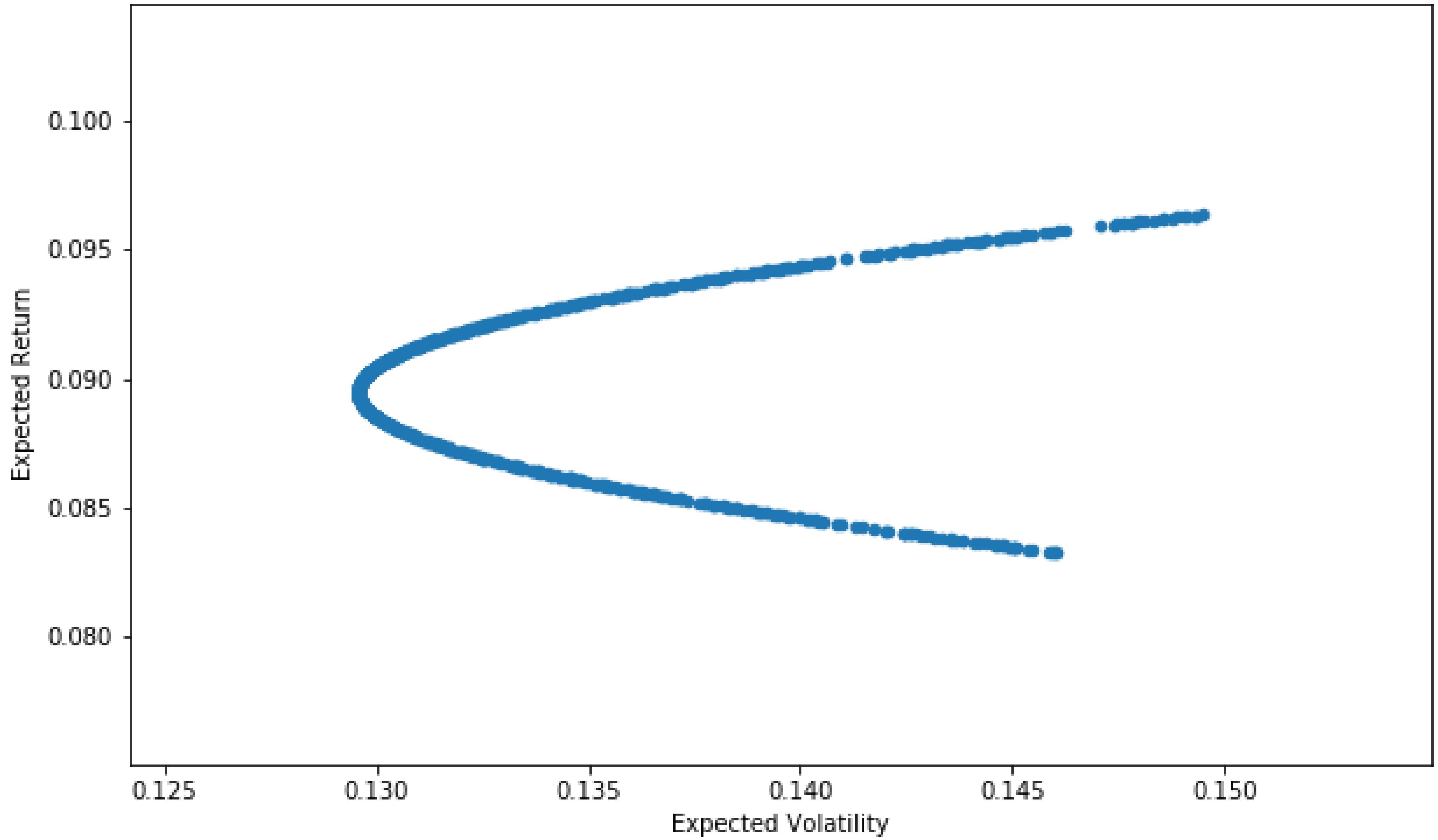
	Return	Volatility
0	0.092203	0.132899
1	0.092120	0.132703
2	0.083389	0.145347
3	0.091460	0.131344
4	0.083628	0.144191

```
[ ] portfolios.tail()
```

	Return	Volatility
995	0.084510	0.140263
996	0.092904	0.134791
997	0.090293	0.129872
998	0.093707	0.137443
999	0.083807	0.143354

```
[ ] portfolios.plot(x='Volatility', y='Return', kind='scatter', figsize=(10, 6));
    plt.xlabel('Expected Volatility')
    plt.ylabel('Expected Return')
```

```
Text(0, 0.5, 'Expected Return')
```



Beta
One of the main pillars of the CAPM

Beta

- Beta helps us quantify the relationship between a security and the overall market portfolio

Recession (Crisis) vs Flourishing Economy

- Recession (Crisis)

- Flourishing

- Safer stocks: these stocks will earn less than the market portfolio when the economy grows
- Riskier stocks: stocks that will earn more than the market portfolio when the economy grows

Measuring Beta

- Beta: measures the market risk that cannot be avoided through diversification

$$\beta = \frac{Cov\,(r_x, r_m)}{\sigma_m^2}$$

- The relationship between a stock and the market:

No relationship	Defensive	Aggressive
$\beta = 0$	$\beta < 1$	$\beta > 1$

Examples

Some companies are less dependent on the economic cycle. Walmart's clients will continue to buy food and household products.

$\beta = 0.09$

Other companies like Ford are impacted more significantly. In times of recession people are unwilling to buy new cars and are postpone such purhcases

$\beta = 1.1$

Quiz

For which value of beta is a stock considered to have no relationship with the market?

- Beta < 0
- Beta = 0 (correct)
- Beta < 1
- Beta > 1

Pratice

Lab06_04_Calculating the Beta of a Stock

```
import numpy as np
import pandas as pd
from pandas_datareader import data as wb
import yfinance as yf
yf.pdr_override()

tickers = ['PG', '^GSPC']
data = pd.DataFrame()
for t in tickers:
    data[t] = wb.DataReader(t, start='2012-1-1', end='2016-12-31')['Adj Close']
```

```
[ ] sec_returns = np.log( data / data.shift(1) )
```

```
[ ] sec_returns = np.log( data / data.shift(1) )
```

```
cov = sec_returns.cov() * 250
cov
```

	PG	^GSPC
PG	0.020409	0.010078
^GSPC	0.010078	0.016362

```
[ ] cov_with_market = cov.iloc[0,1]
    cov_with_market
```

```
0.010078075182705783
```

```
[ ] market_var = sec_returns['^GSPC'].var() * 250
    market_var
```

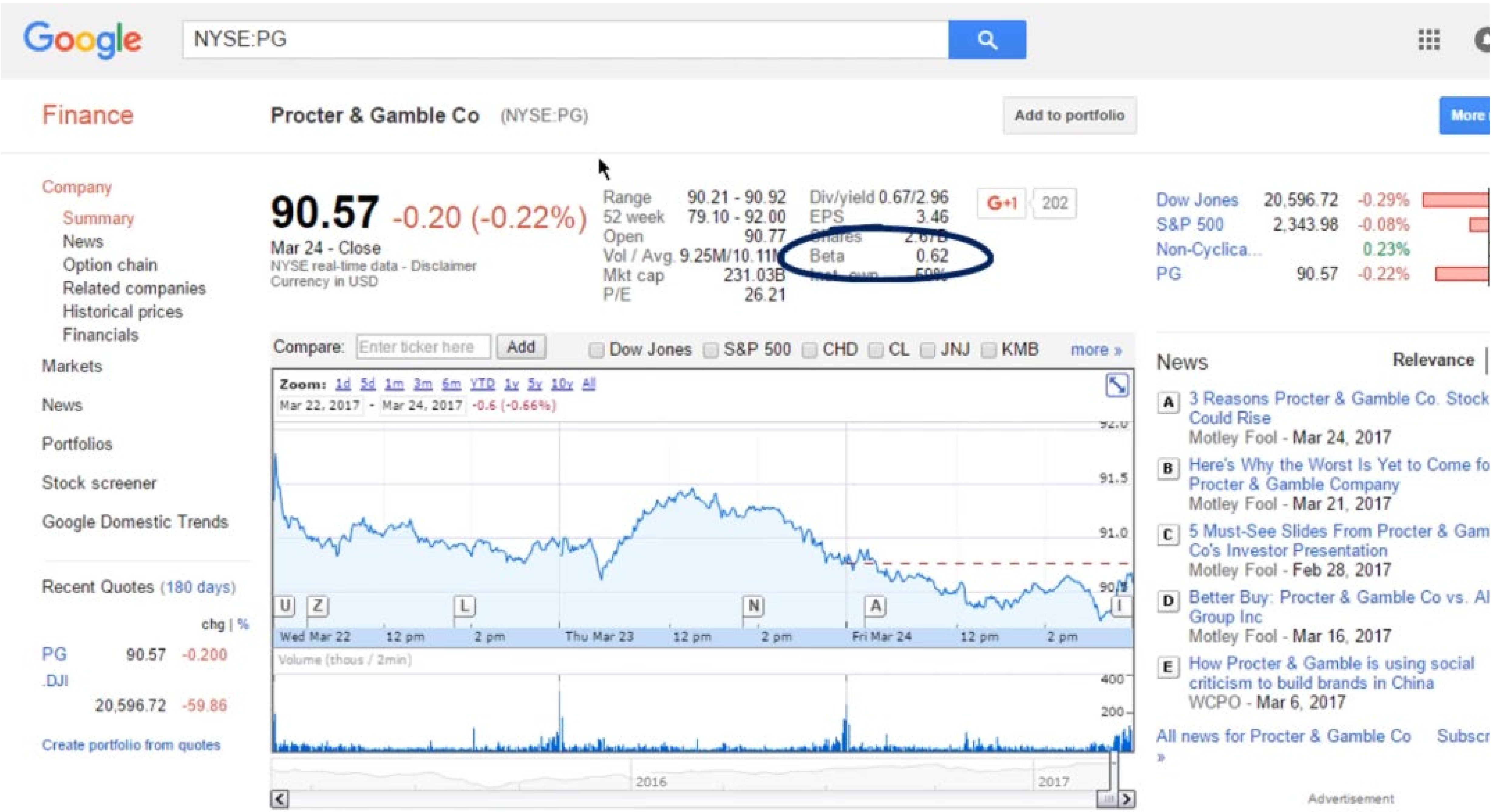
```
0.016361631002308474
```

** Beta: **

$$\beta_{pg} = \frac{\sigma_{pg,m}}{\sigma_m^2}$$

```
[ ]  PG_beta = cov_with_market / market_var
     PG_beta
```

```
0.6159578578250458
```


Google
NYSE:PG
Finance
Procter & Gamble Co (NYSE:PG)
Add to portfolio
More
Company
Summary
News
Option chain
Related companies
Historical prices
Financials
Markets
News
Portfolios
Stock screener
Google Domestic Trends
Recent Quotes (180 days)
chg | %
PG 90.57 -0.200
.DJI 20,596.72 -59.86
Create portfolio from quotes
90.57 -0.20 (-0.22%)
Mar 24 - Close
NYSE real-time data - Disclaimer
Currency in USD
Range 90.21 - 90.92
52 week 79.10 - 92.00
Open 90.77
Vol / Avg. 9.25M/10.11M
Mkt cap 231.03B
P/E 26.21
Div/yield 0.67/2.96
EPS 3.46
Beta 0.62
G+1 202
Dow Jones 20,596.72 -0.29%
S&P 500 2,343.98 -0.08%
Non-Cyclica... 0.23%
PG 90.57 -0.22%
Compare: Enter ticker here Add
Dow Jones S&P 500 CHD CL JNJ KMB more »
Zoom: 1d 5d 1m 3m 6m YTD 1y 5y 10y All
Mar 22, 2017 - Mar 24, 2017 -0.6 (-0.66%)
91.5
91.0
Wed Mar 22 12 pm 2 pm Thu Mar 23 12 pm 2 pm Fri Mar 24 12 pm 2 pm
Volume (thous / 2min)
400
200
2016
2017
News
Relevance
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B Here's Why the Worst Is Yet to Come fo Procter & Gamble Company
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Capital Asset Pricing Model

Capital Asset Pricing Model

Equirty risk premium

$$\boxed{r_i = r_f + \beta_{im}(r_m - r_f)}$$

r_f - risk-free

β_{im}- beta between the stock and the market

r_m - market return

CAPM in Pratice

Risk-free	Beta	Equity Risk Premium
Approximate with 10-year US government bond yield: 2.5%	Approximate the market portfolio with the S&P500: 0.62	Historically, it has been between 4.5% and 5.5%

$$\boxed{r_i = r_f + \beta_{im}(r_{m-}r_f)}$$

r(i) = 2.5% + 0.62*5%=5.6%

Quiz

According to CAPM, which of the following is not relevant for the calculation of the expected return of a stock?

- The variance of the market
- The risk-free rate
- The risk premium
- The variance of the stock (corect)

Pratice

Lab06_05_Calculating the Expected Return of a Stock (CAPM)

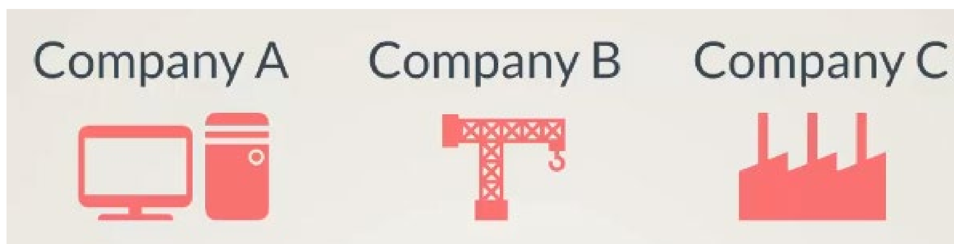
Sharpe Ratio

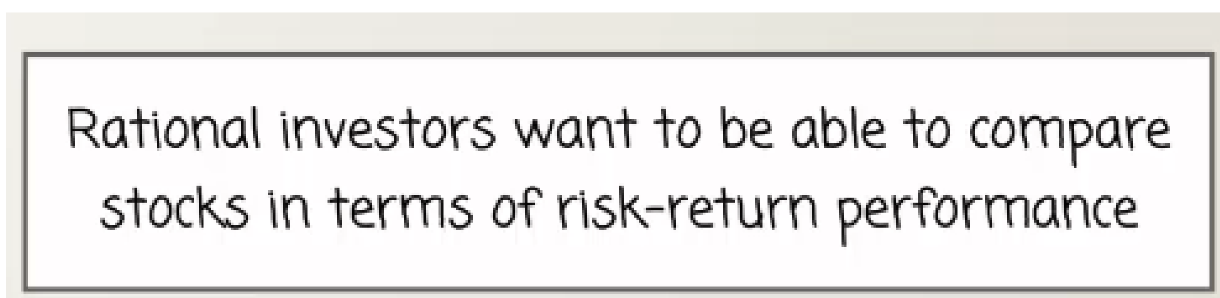
Sharpe Ratio

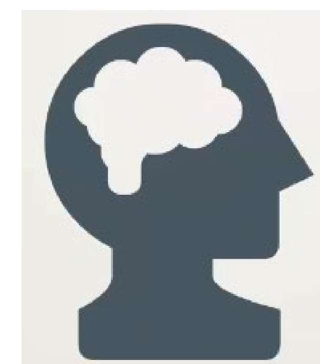
- A rationa investor: consider both risk and return

Rational investors want to maximize their returns

and minimize their risk hence invest in less volatile securities

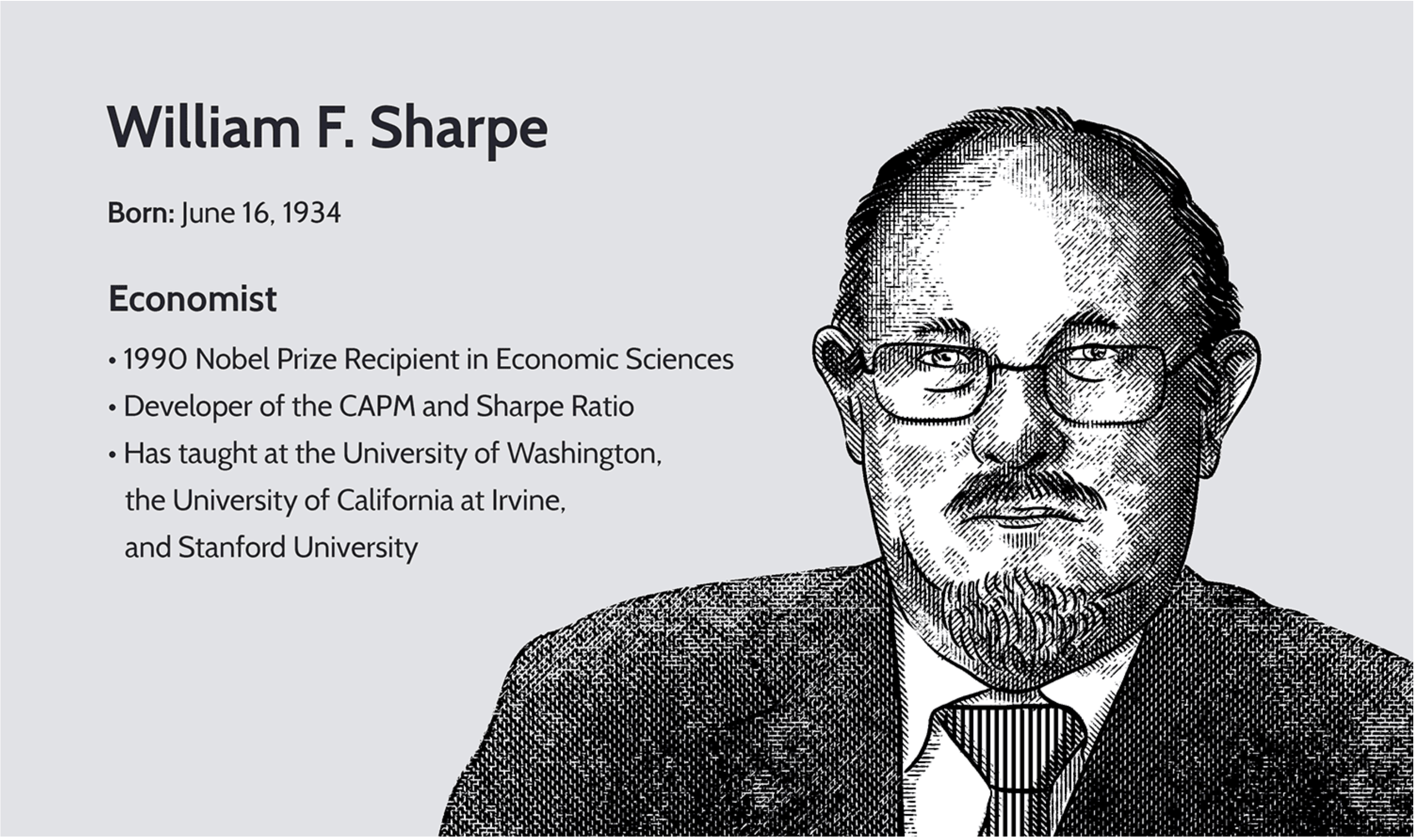




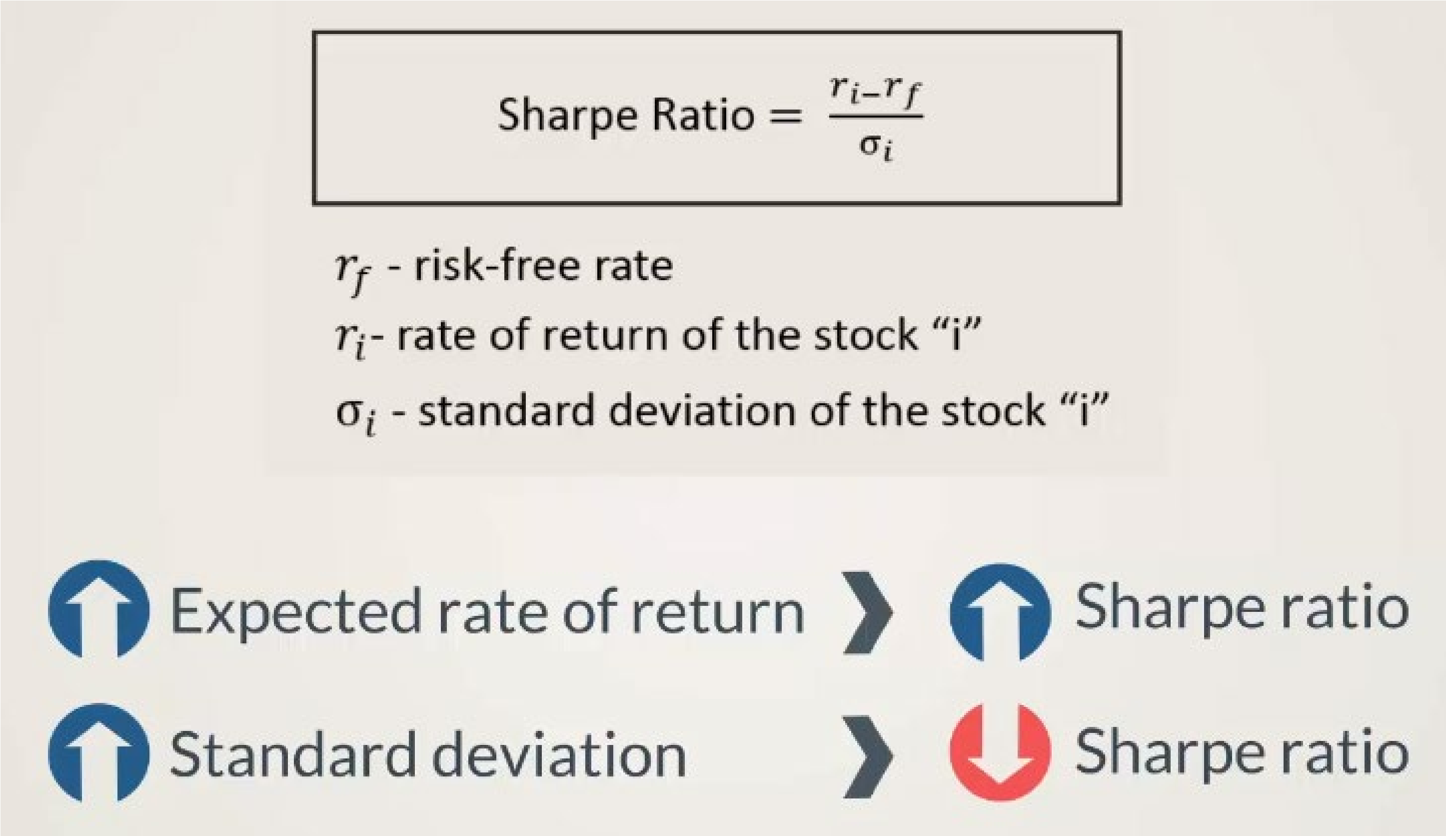


Sharpe Ratio

- William Sharpe came up with the Sharpe Ratio



Sharpe Ratio



- The Sharpe Ratio allows us to compare:

- [x] Stock A vs. Stock B
- [x] Investment Portfolio A vs. Investment Portfolio B

Quiz

Which of the following values is irrelevant for the calculation of Sharpe ratio?

- The risk-free rate
- The excess return of the stock
- The variance of the market (correct)
- The standard deviation of the stock

Pratice

Lab06_05

Calculate the expected return of P&G (CAPM):

$$\overline{r_{pg}} = r_f + \beta_{pg}(\overline{r_m} - r_f)$$

```
[ ] PG_er = 0.025 + PG_beta * 0.05
    PG_er
```

```
0.05579789289125229
```

Alpha

Alpha

- Interpreting alpha: a measure of how good or bad an investment manager is doing

Compensation for risk

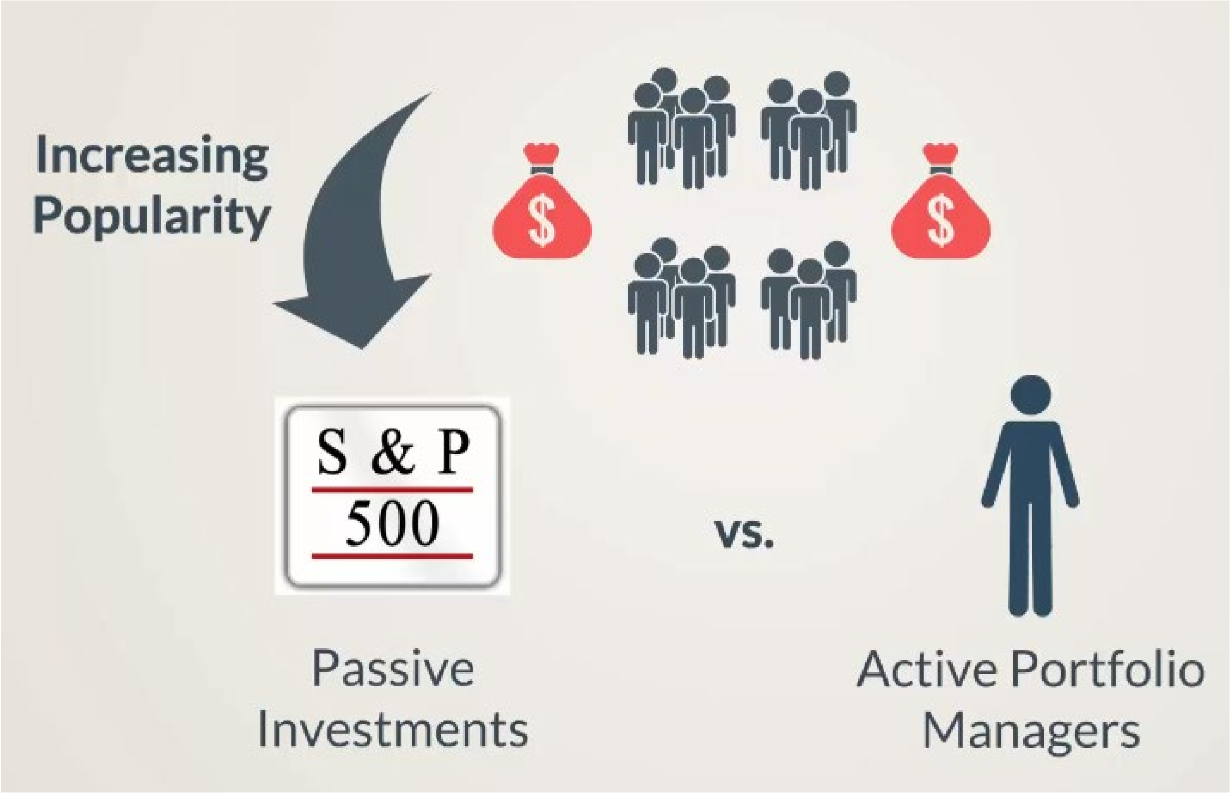
$$r_i = \alpha + r_f + \beta_{im}(r_m - r_f)$$

The standard CAPM setting assumes an alpha equal to 0

- How does an investment manager ty to outperform the market?
  - Industry knowledge, Company-specific information, trading know-how
  - **A good portfolio** manager: outperforms the market and earns a positive alpha
  - **A poor portfolio** manager: underperforms the market and earns a negative or zero alpha

Investment Trend

- **Passive trading**: handpicking successful investments; long-term investments
- **Active trading**: adjuisting investment portfolios on a frequent basis



Quiz

What does alpha show us?

- How to outperform the market
- How much return we get without bearing extra risk (correct)
- How to gain trading know-how
- How to compare investments with divergent risk profiles

Pratice

Lab06_06_Estimating the Sharpe Ratio

Sharpe ratio:

$$Sharpe = \frac{\overline{r_{pg}} - r_f}{\sigma_{pg}}$$

```
[ ] Sharpe = (PG_er - 0.025) / (sec_returns['PG'].std() * 250 ** 0.5)
    Sharpe
```

```
0.21557986697329162
```

THANKS FOR YOUR ATTENTION!